Pure Y1 Ch 14 Exponentials and Logarithms

1) Given that $x = 3$ and $y = -1$, evaluate these expressions without a calculator.

a) 5^4 125

b) $3^{\frac{1}{3}y}$

c) 2^{2x-1} 32

d) 7^{1-y} 49

e) 11^{x+3y}

2) Simplify these expressions, writing each answer as a single power.

a) $6^8 \div 6^2$ 6^6

b) $y^3 \times (y^9)^2$ y^{21}

c) $\frac{2^6 \times 2^9}{2^8}$ 2^7

d) $\sqrt{x^8}$ x^4

3) Plot the following data on a scatter graph and draw a line of best fit. Determine the gradient and intercept of the line of best fit, giving the answers to one decimal place.

gradient 1.5, intercept 4.1

x	1.2	2.1	3.5	4	5.8
y	5.8	7.4	9.4	10.3	12.8

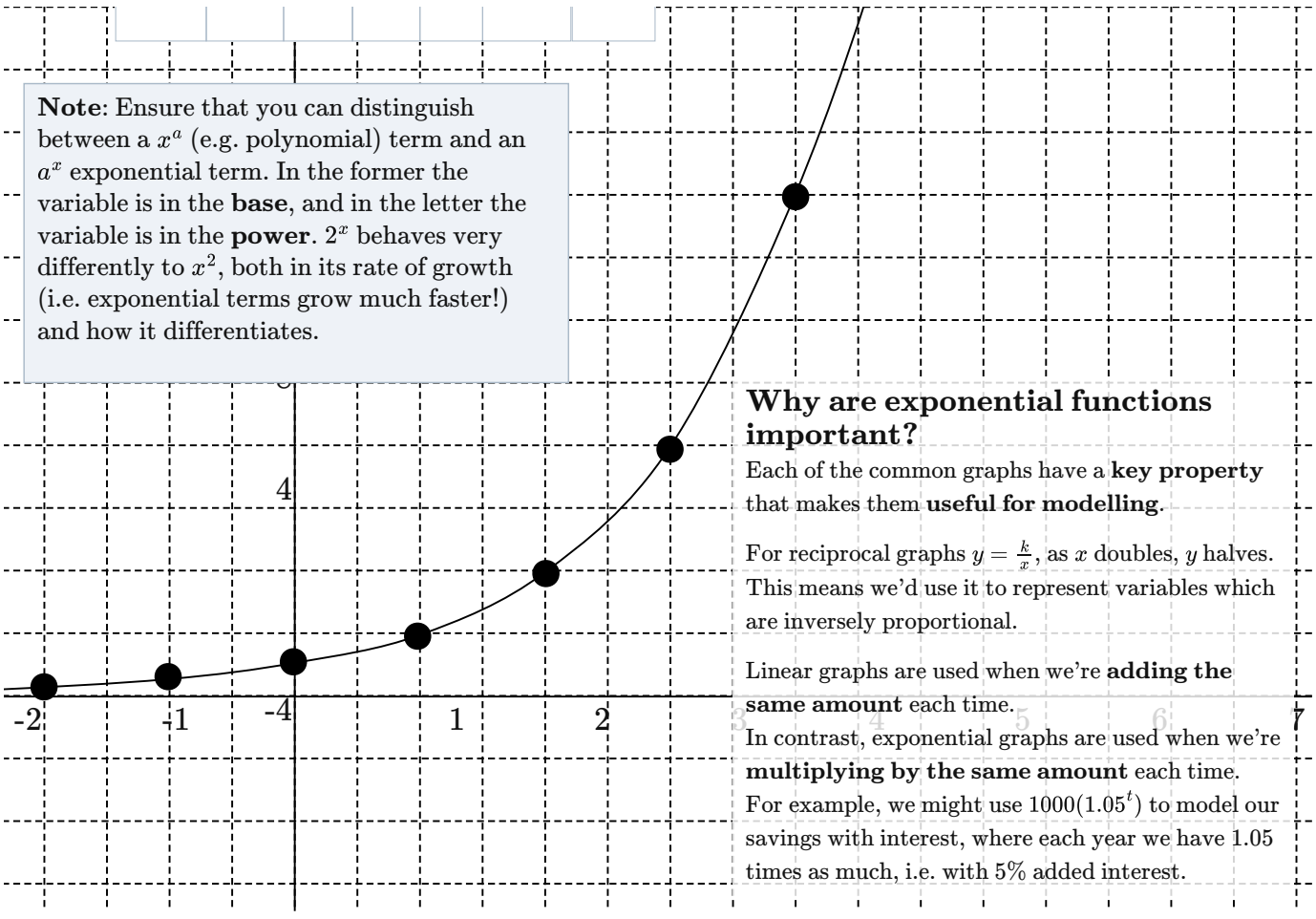
1: Sketch exponential graphs.

2: Use an interpret models that use exponential functions.

3: Be able to differentiate e^{kx}.

4: Understand the log function and use laws of logs.

5: Use logarithms to estimate values of constants in non-linear models.



Note: Ensure that you can distinguish between a x^a (e.g. polynomial) term and an a^x exponential term. In the former the variable is in the **base**, and in the letter the variable is in the **power**. 2^x behaves very differently to x^2, both in its rate of growth (i.e. exponential terms grow much faster!) and how it differentiates.

Why are exponential functions important?

Each of the common graphs have a **key property** that makes them **useful for modelling**.

For reciprocal graphs $y = \frac{k}{x}$, as x doubles, y halves. This means we'd use it to represent variables which are inversely proportional.

Linear graphs are used when we're **adding the same amount** each time.
In contrast, exponential graphs are used when we're **multiplying by the same amount** each time.
For example, we might use $1000(1.05^t)$ to model our savings with interest, where each year we have 1.05 times as much, i.e. with 5% added interest.

Contrasting exponential graphs 5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

Graph Transformations 6

Sketch $y = 2^{x+3}$

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

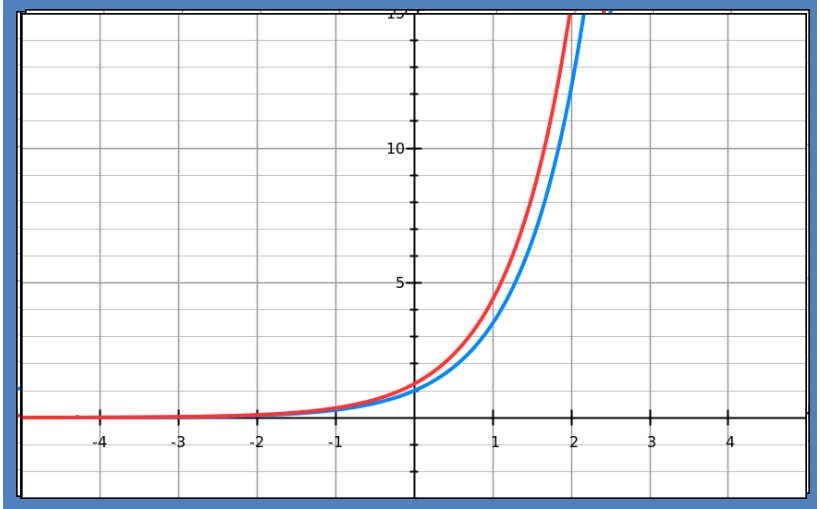
a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

Click

Function	Gradient	
$y = 1^x$	$\frac{dy}{dx} = 0$	>
$y = 1.5^x$	$\frac{dy}{dx} = 0.41 \times 1.5^x$	>
$y = 2^x$	$\frac{dy}{dx} = 0.69 \times 2^x$	>
$y = 2.5^x$	$\frac{dy}{dx} = 0.92 \times 2.5^x$	>
$y = 3^x$	$\frac{dy}{dx} = 1.10 \times 3^x$	>
$y = 3.5^x$	$\frac{dy}{dx} = 1.25 \times 3.5^x$	>

Compare each exponential function against its respective gradient function. What do you notice?



Function	Gradient	
$y = 1^x$	$\frac{dy}{dx} = 0$	>
$y = 1.5^x$	$\frac{dy}{dx} = 0.41 \times 1.5^x$	>
$y = 2^x$	$\frac{dy}{dx} = 0.69 \times 2^x$	>
$y = 2.5^x$	$\frac{dy}{dx} = 0.92 \times 2.5^x$	>
$y = 3^x$	$\frac{dy}{dx} = 1.10 \times 3^x$	>
$y = 3.5^x$	$\frac{dy}{dx} = 1.25 \times 3.5^x$	>

$y = 2.5^x$ and $y = 3^x$ seem to be similar to their respective gradient functions. So is there a base between 2.5 and 3 where the **function is equal to its gradient function**?

$e = 2.71828...$ is known as **Euler's Number.**

It is one of the five most fundamental constants in mathematics (0, 1, i, e, π).

It has the property that:

$$y = e^x \quad \rightarrow \quad \frac{dy}{dx} = e^x$$

Although any function of the form $y = a^x$ is known as **an** exponential function, e^x is known as "**the**" exponential function.

You can find the exponential function on your calculator, to the right (above the "ln" key)

If $y = e^{kx}$, where k is a constant, then $\frac{dy}{dx} = ke^{kx}$

Note: This is not a standalone rule but an application of something called the '*chain rule*', which you will encounter in Year 2.

Different e^{5x} with respect to x.

Different e^{-x} with respect to x.

Different $4e^{3x}$ with respect to x.

Note: In general, when you scale the function, you scale the derivative/integral.

Sketch $y = e^{3x}$

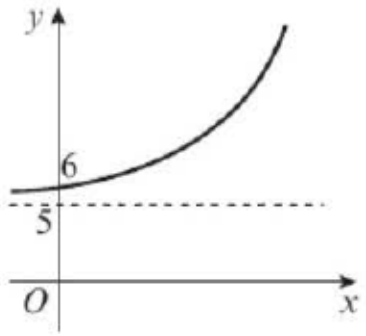
Sketch $y = 5e^{-x}$

Sketch $y = 2 + e^{\frac{1}{3}x}$

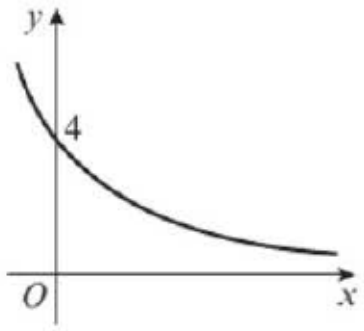
Sketch $y = e^{-2x} - 1$

4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.

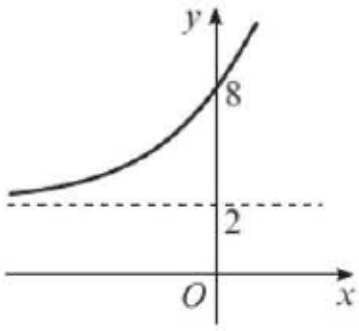
a



b



c



6 Differentiate the following with respect to x.

a e^{6x} **b** $e^{-x/3}$ **c** $7e^{2x}$ **d** $5e^{0.4x}$ **e** $e^{3x} + 2e^{x}$ **f** $e^{x}(e^{x} + 1)$

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

8 The function f is defined as $f(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

Where does e come from, and why is it so important?

$e = 2.71828\ldots$
is known as **Euler's Number**, and is considered one of the five fundamental constants in maths:
$0, \quad 1, \quad \pi, \quad e, \quad i$

Its value was originally encountered by Bernoulli who was solving the following problem:
You have £1. If you put it in a bank account with 100% interest, how much do you have a year later? If the interest is split into 2 instalments of 50% interest, how much will I have? What about 3 instalments of 33.3%? And so on…

No.	Money after a year
1	$1 \times 2^1 = £2$
2	$1 \times 1.5^2 = £2.25$
3	$1 \times 1.\dot{3}^3 = £2.37$
4	$1 \times 1.25^4 = £2.44$
n	$\left(1 + \frac{1}{n}\right)^n$

As n becomes larger, the amount after a year approaches £2.71…, i.e. e!

Thus:
$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$

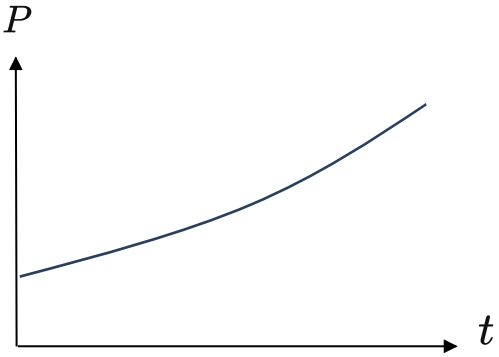
But we have seen that differentiation by first principles uses 'limits'. It is therefore possible to prove from the definition above that $\frac{d}{dx}(e^x) = e^x$, and **these two definitions of e are considered to be equivalent***.

e therefore tends to arise in problems involving limits, and also therefore crops up all the time in anything involving differentiation and integration. Let's see some applications…

There are two key features of exponential functions which make them suitable for **population growth**:

1. **a^x gets a times bigger each time x increases by 1. (Because $a^{x+1}=a\times a^x$)**
   With population growth, we typically have a fixed percentage increase each year. So suppose the growth was 10% a year, and we used the equivalent decimal multiplier, 1.1, as a. Then 1.1^t, where t is the number of years, would get 1.1 times bigger each year.
2. **The rate of increase is proportional to the size of the population at a given moment.**
   This makes sense: The 10% increase of a population will be twice as large if the population itself is twice as large.

Suppose the population P in *The Republic of Nqmir* is modelled by $P = 100e^{3t}$ where t is the numbers years since *The Republic* was established.



What is the initial population?

What is the initial rate of population growth?

"Use of Technology" I use the $e^{\square}$ key. I can also use [ALPHA] [e] to get e without a power.

[Textbook] The density of a pesticide in a given section of field, P mg/m^2, can be modelled by the equation $P = 160e^{-0.006t}$
where t is the time in days since the pesticide was first applied.
a. Use this model to estimate the density of pesticide after 15 days.
b. Interpret the meaning of the value 160 in this model.
c. Show that $\frac{dP}{dt} = kP$, where k is a constant, and state the value of k.
d. Interpret the significance of the sign of your answer in part (c).
e. Sketch the graph of P against t.

a

In general, the 'initial value', when $t = 0$, is the coefficient of the exponential term.

b

$t = 0 \quad P = 160$

$\frac{dP}{dt} = (160 \times -0.006)e^{-0.006t}$

c

d

5 On Earth, the atmospheric pressure, p, in bars can be modelled approximately by the formula $p = e^{-0.13h}$ where h is the height above sea level in kilometres.

a Use this model to estimate the pressure at the top of Mount Rainier, which has an altitude of $4.394\,\text{km}$.
b Demonstrate that $\frac{\mathrm{d}p}{\mathrm{d}h} = kp$ where k is a constant to be found.
c Interpret the significance of the sign of k in part **b**.
d This model predicts that the atmospheric pressure will change by $s\%$ for every

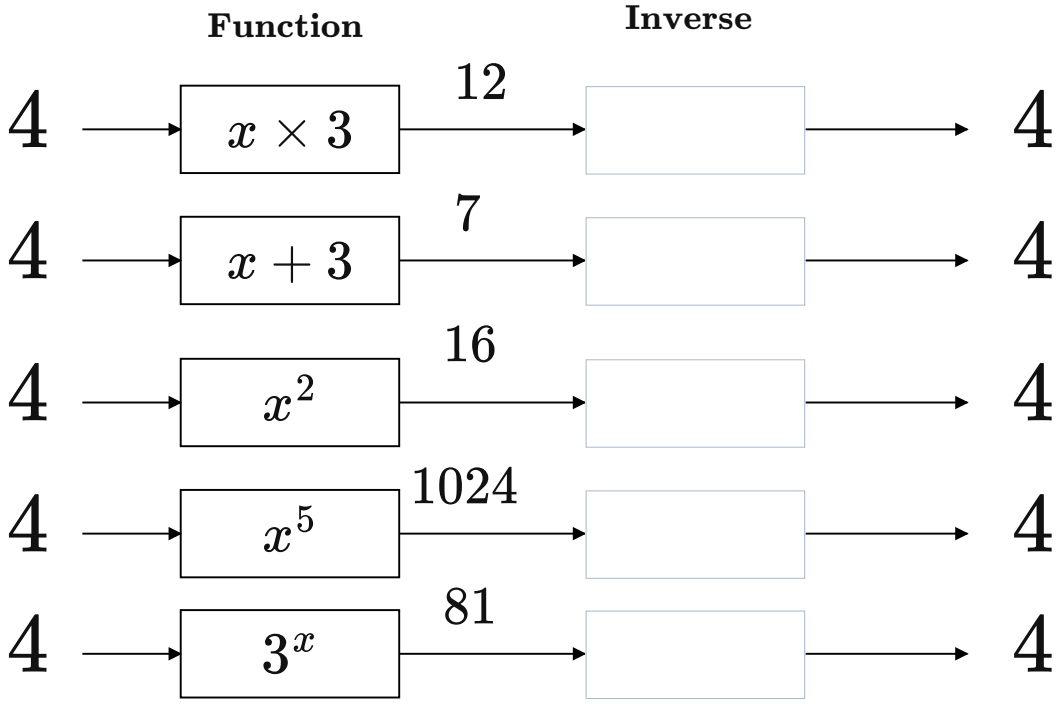
Km gained in height. Calculate the value of s.

6 Nigel has bought a tractor for £20 000. He wants to model the depreciation of the value of his tractor, £T, in t years. His friend suggests two models:

Model 1: $T = 20\,000e^{-0.24t}$

Model 2: $T = 19\,000e^{-0.255t} + 1000$ **a** Use both models to predict the value of the tractor after one year. Compare your results.
b Use both models to predict the value of the tractor after ten years. Compare your results.
c Sketch a graph of T against t for both models.
d Interpret the meaning of the 1000 in model 2, and suggest why this might make model 2 more realistic. (1 mar

You know the inverse of many mathematical operations; we can undo an addition by 2 for example by subtracting 2. But is there an inverse function for an exponential function?



Such functions are known as **logarithms**, and exist in order to provide an inverse to exponential functions.

$\log_a n$ ("said log base a of n") is equivalent to $a^x = n$.
The log function outputs the **missing power**.

$$3^2 = 9 \iff \log_3 9 = 2$$

Here are two methods of interchanging between these forms.
Pick your favourite!

Method 1: 'Missing Power'

- Note first the base of the log must match the base of the exponential.
- $\log_2 8$ for example asks the question "2 to **what power** gives 8?"

We can imagine inserting the output of the log just after the base. Click the button!

$\log_2 8 = 3$

Click to start Animation

Method 2: Do same operation to each side of equation.

Since KS3 you're used to the idea of doing the same thing to each side of the equation that 'undoes' whatever you want to get rid of.

$$3x + 2 = 11$$
$$\overset{(-2)}{\quad} \quad 3x = 9 \quad \overset{(-2)}{\quad}$$

"log base a" undoes "a to the power of" and vice versa, as they are inverse functions.

$$\log_2 8 = 3$$
$$\overset{(2^{\square})}{\quad} \quad 8 = 2^3 \quad \overset{(2^{\square})}{\quad}$$

$\log_5 25 = 2$ ← Think: "5 to the power of what gives you 25?"

$\log_3 81 = 4$

$\log_2 32 = 5$

$\log_{10} 1000 = 3$

$\log_4 1 = 0$ ✎ $\log_a 1 = 0$ for all a.

$\log_4 4 = 1$

$\log_2 \left(\frac{1}{2}\right) = -1$

$\log_3 \left(\frac{1}{27}\right) = -3$

$\log_2 \left(\frac{1}{16}\right) = -4$

$\log_a \left(a^3\right) = 3$

$\log_4 (-1) =$

There are three buttons on your calculator for computing logs:

$\log_{\square}\square$

$\log_3 7 = \mathbf{1.771}$

$\log_5 0.3 = \mathbf{-0.7}$

ln

$\ln 10 = \mathbf{2.3025}$

$\ln e = \mathbf{1}$

ln is the "**natural log of x**", meaning "log to the base e", i.e. it the inverse of e^x.

$$\ln(x) = \log_e(x)$$

We will use it more extensively later this chapter.

log

$\log 100 = \mathbf{2}$

Just like the $\sqrt{}$ symbol without a number is $\sqrt[2]{\square}$ by default, log without a base is **base 10** by default when used on your calculator (although confusingly "log" can mean "ln" in mathematical papers)

6 a Without using a calculator, justify why the value of $\log_2 50$ must be between 5 and 6.

b Use a calculator to find the exact value of $\log_2 50$ to 4 significant figures.

7 a Find the values of:

i $\log_2 2$ **ii** $\log_3 3$ **iii** $\log_{17} 17$ **b** Explain why $\log_a a$ has the same value for all positive values of a ($a \neq 1$).

8 a Find the values of:

i $\log_2 1$ **ii** $\log_3 1$ **iii** $\log_{17} 1$ **b** Explain why $\log_a 1$ has the same value for all positive values of a ($a \neq 1$).

1

Non-calculator!

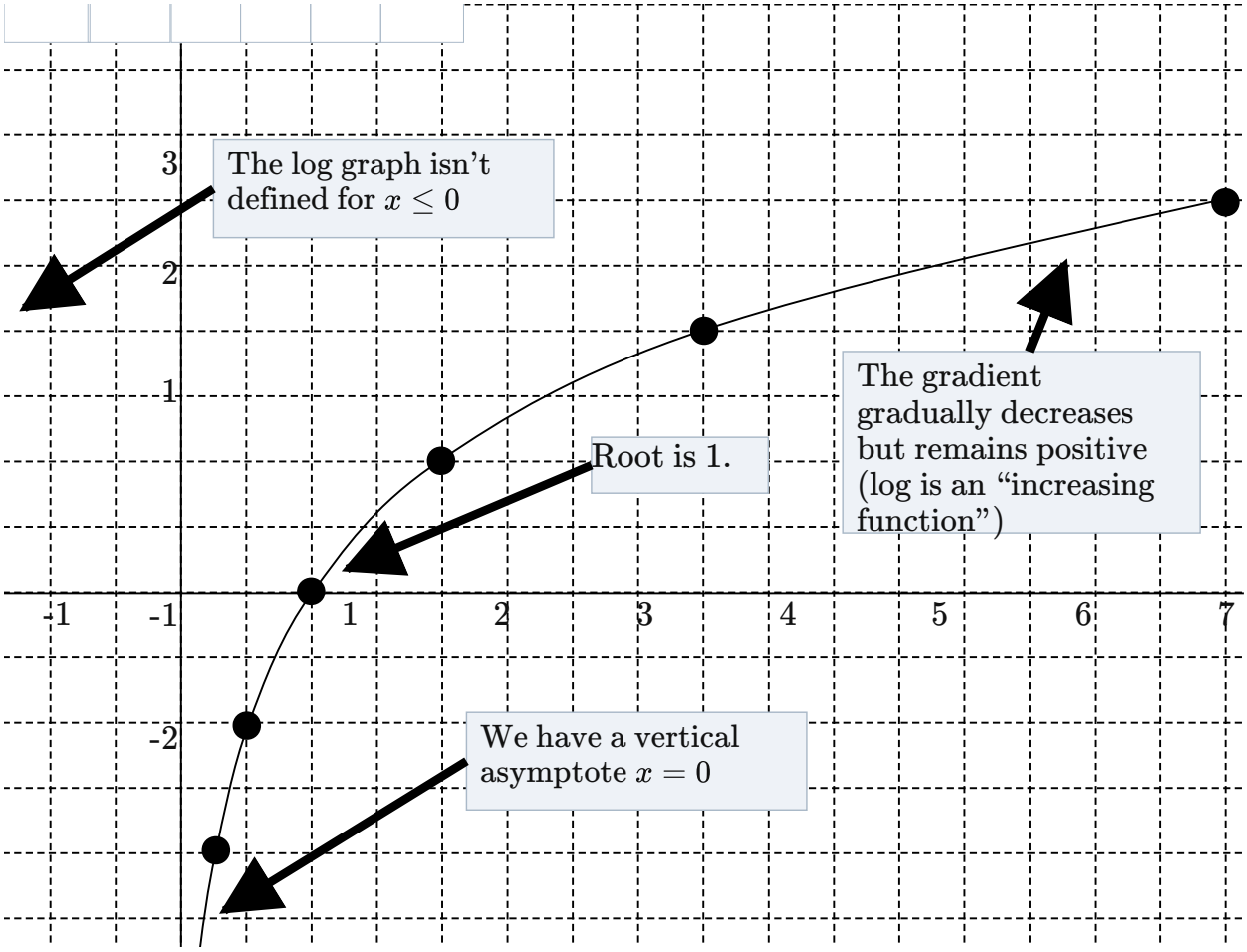
2

[MAT 2013 1F] Three *positive* numbers a, b, c satisfy
$\log_b a = 2$
$\log_b (c-3) = 3$
$\log_a (c+5) = 2$
This information:

A) specifies a uniquely;
B) is satisfied by two values of a;
C) is satisfied by infinitely many values of a;
D) is contradictory

If we take exponents of the three equations:

$$\mathbf{a=b^2,\ \ c-3=b^3,\ \ c+5=a^2}$$



Three main laws:

$$\log_a x + \log_a y = \log_a xy$$

$$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$$

$$\log_a \left(x^k\right) = k\log_a x$$

Special cases:

$$\log_a a = 1 \quad (a > 0, a \neq 1)$$

$$\log_a 1 = 0 \quad (a > 0, a \neq 1)$$

$$\log \left(\frac{1}{x}\right) = \log \left(x^{-1}\right) = -\log (x)$$

Not in syllabus (but in MAT/PAT):

$$\log_a b = \frac{\log_c b}{\log_c a}$$

The logs must have a consistent base.

i.e. You can move the power to the front.

We often try to avoid leaving fractions inside logs. So if the answer was:

$$\log_2 \left(\frac{1}{3}\right)$$

You should write your answer as: $-\log_2 3$

Reciprocating the input negates the output.

This is known as **changing the base**.

So to get $\log_2 9$ in terms of log base 3:

$$log_2 9 = \frac{log_3 9}{log_3 2} = \frac{2}{log_3 2}$$

Write as a single logarithm:

a. $\log_3 6 + \log_3 7$

b. $\log_2 15 - \log_2 3$

c. $2\log_5 3 + 3\log_5 2$

d. $\log_{10} 3 - 4\log_{10}\left(\frac{1}{2}\right)$

a

b

c

d

Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$

a. $\log_a(x^2yz^3)$

b. $\log_a\left(\frac{x}{y^3}\right)$

c. $\log_a\left(\frac{x\sqrt{y}}{z}\right)$

d. $\log_a\left(\frac{x}{a^4}\right)$

a

b

c

d

These are **NOT LAWS OF LOGS**, but are mistakes students often make:

$$\log_a(b+c) = \log_a b + \log_a c$$

✗ FAIL

$$(\log_2 x)^3 = 3\log_2 x$$

✗ FAIL

Solve the equation $\log_{10}4 + 2\log_{10}x = 2$

This is a very common type of exam question.
The strategy is to **combine the logs into one** and isolate on one side.

Given that $2\log_2(x+15) - \log_2 x = 6$,

(a) show that $x^2 - 34x + 225 = 0$. (5)

(b) Hence, or otherwise, solve the equation $2\log_2(x+15) - \log_2 x = 6$. (2)

a

$\log_2(x$
$\log_2\left($
$2^6 = \frac{(x+1}{x}$
$64x = (x$
$64x = x^2$
$x^2 - 34x$

$(x - 25)($
$x = 25 or x = 9$

b

5 a Given that $\log_3(x+1) = 1 + 2\log_3(x-1)$, show that $3x^2 - 7x + 2 = 0$.

b Hence, or otherwise, solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

6 Given that a and b are positive constants, and that $a > b$, solve the simultaneous equations

$a + b = 13 \log_6 a + \log_6 b = 2$

Challenge:
By Writing $\log_a x = m$ and $\log_a y = n$
Prove that $\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$

These are all strictly **non-calculator**!

1 [AEA 2010 Q1b] Solve the equation

$$\log_3(x-7) - \tfrac{1}{2}\log_3 x = 1 - \log_3 2$$

Solution: x=$\frac{49}{4}$

2 Anna, who is confused about the rules of logarithms, states that

$(\log_3 p)^2 = \log_3(p^2)$

$\log_3(p+q) = \log_3 p + \log_3 q$

However, there is a value for p and a value for q for which both statements are correct. Find their values.

First equation:

3 [MAT 2007 1I] Given that a and b are positive and

$$4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$$

what is the greatest possible value of a?

4 [MAT 2002 1F] Observe that $2^3 = 8$, $2^5 = 32$, $3^2 = 9$ and $3^3 = 27$. From these facts, we can deduce that $\log_2 3$ is:

A) between $1\frac{1}{3}$ and $1\frac{1}{2}$

B) between $1\frac{1}{2}$ and $1\frac{2}{3}$

C) between $1\frac{2}{3}$ and 2

D) none of the above

Suppose that $\log_2 3 > \frac{3}{2}$. Taking 2 to the power of

Solve $3^x = 20$

Solve $5^{4x-1} = 61$

Solve $3^x = 2^{x+1}$

Test Your Understanding 38

1 Solve $3^{2x-1} = 5$, giving your answer to 3dp.

2 Solve $3^{x+1} = 4^{x-1}$, giving your answer to 3dp.

3 Solve $2^x 3^{x+1} = 5$, giving your answer in exact form.

3 Solve the following equations, giving your answers to 3 significant figures where appropriate.

a $3^{x+1} = 2000$ (2 mar

b $\log_5(x-3) = -1$ (2 mar

4 a Sketch the graph of $y = 4^x$, stating the coordinates of any points where the graph crosses the axes.

b Solve the equation $4^{2x} - 10(4^x) + 16 = 0$.

1 **EXTENSION** [MAT 2011 1H] How many *positive* values x which satisfy the equation:

$$x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + \log_{0.5} 0.25$$

2 [MAT 2013 1J] For a real number x we denote by $[x]$ the largest integer less than or equal to x. Let n be a natural number. The integral

$$\int_0^n [2^x] dx$$

equals:

(A) $\log_2((2^n - 1)!)$
(B) $n2^n - \log_2((2^n)!)$
(C) $n2^n$
(D) $\log_2((2^n)!)$

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.
We also saw that e^x is "**the**" exponential function.
The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

✎ The inverse of $y = e^x$ is $y = \ln x$

$$\ln e^x = \mathbf{x}$$
$$e^{\ln x} = \mathbf{x}$$

Since "e to the power of" and "ln of" are inverse functions, they cancel each other out!

Solve $e^x = 5$

Solve $2\ln x + 1 = 5$

In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation **is** then quadratic, or (strongly recommended!) **go straight for the factorisation**.

Solve $e^{2x} + 2e^x - 15 = 0$

Solve $e^x - 2e^{-x} = 1$

Solve $\ln(3x+1) = 2$

Solve $e^{2x} + 5e^{x} = 6$

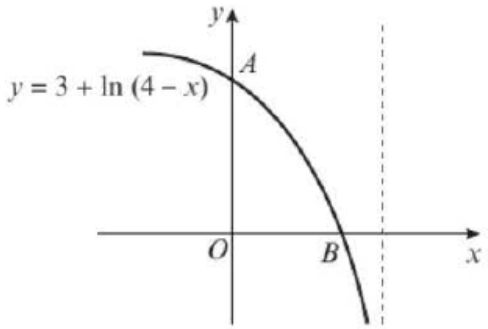
Solve $2^{x}e^{x+1} = 3$ giving your answer as an exact value.

4 Find the exact solutions to the equation $e^{x} + 12e^{-x} = 7$.

6 Solve $3^{x}e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$

8 The graph of $y = 3 + \ln(4 - x)$ is shown to the right.

a State the exact coordinates of point A.
b Calculate the exact coordinates of point B.

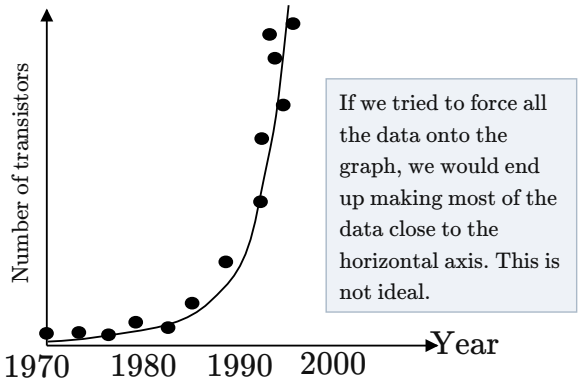


Challenge: The graph of the function passes through (0,5) and (6,10). Given that the line $y = 2$ is an asymptote to the graph, show that $B = \frac{1}{6}\ln\left(\frac{8}{3}\right)$

In Science and Economics, **experimental data often has exponential growth**, e.g. bacteria in a sample, rabbit populations, energy produced by earthquakes, my Twitter followers over time, etc.

Because exponential functions increase rapidly, it tends to look a bit rubbish if we tried to draw a suitable graph:

Take for example "Moore's Law", which hypothesised that the processing power of computers would double every 2 years. Suppose we tried to plot this for computers we sampled over time:



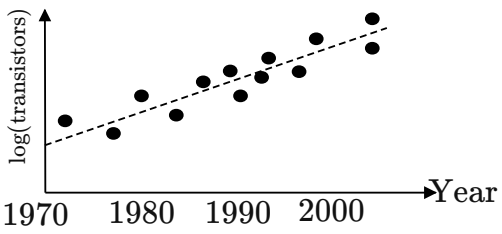
If we tried to force all the data onto the graph, we would end up making most of the data close to the horizontal axis. This is not ideal.

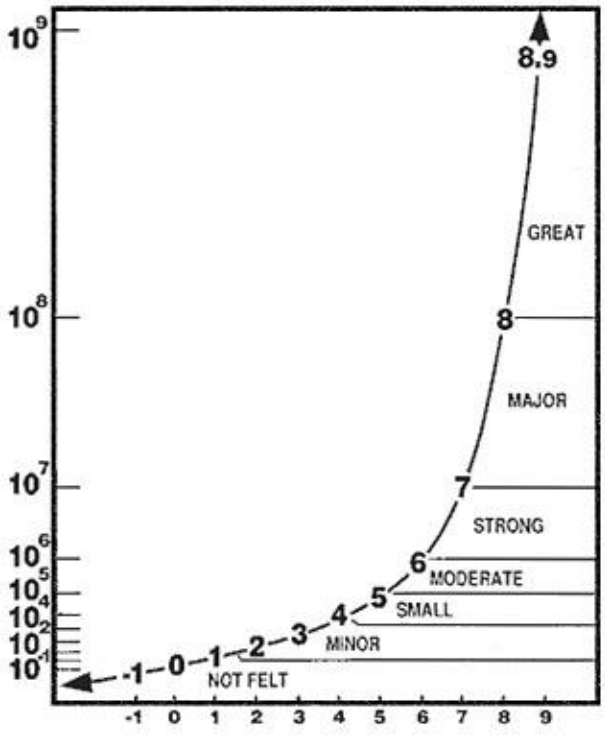
But suppose we **took the log** of the number of transistors for each computer. Suppose the number of transistors one year was y, then doubled 2 years later to get $2y$.

When we log (base 2) these:

$$\begin{aligned} y &\rightarrow \log_2 y \\ 2y &\rightarrow \log_2(2y) = \log_2 2 + \log_2 y \\ &\qquad = 1 + \log_2 y \end{aligned}$$

The logged value only increased by 1! Thus **taking the log of the values turns exponential growth into linear growth (because each time we would have doubled, we're now just adding 1), and the resulting graph is a straight line.**





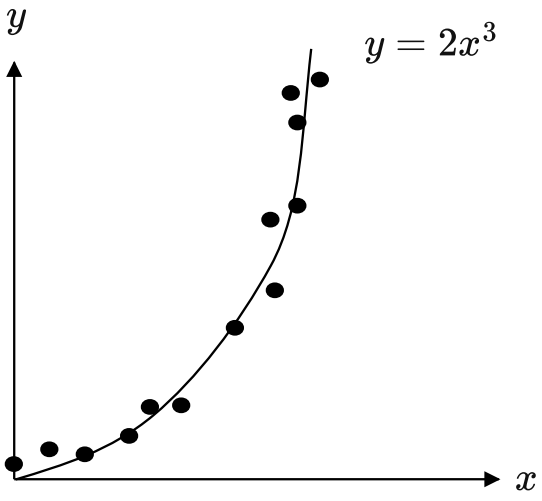
Richter Scale

Because the energy involved in **earthquakes** decreases exponentially from the epicentre of the earthquake, such energy values recorded from different earthquakes would **vary wildly**.

The **Richter Scale** is a **logarithmic scale**, and takes the log (base 10) of the amplitude of the waves, giving a more even spread of values in a more sensible range.

(The largest recorded value on the Richter Scale is 9.5 in Chile in 1960, and 15 would destroy the Earth completely – evil scientists take note)

The result is that an earthquake just 1 greater on the Richter scale would in fact be 10 times as powerful.



We would also have similar graphing problems if we tried to plot data that followed some **polynomial function** such as a quadratic or cubic.

We will therefore look at the process to convert a **polynomial graph into a linear one**, as well as a **exponential graph into a linear one. . .**

Case 1: Polynomial → Linear

Suppose our original model was a polynomial one*:

$$y = ax^n$$

Then taking logs of both sides:

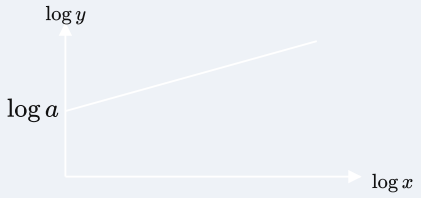
$\log y = \log ax^n$

$\log y = \log a + n \log x$

We can compare this against a straight line:

$$Y = mX + c$$

! If $y = ax^n$, then the graph of $\log y$ against $\log x$ will be a straight line wih gradient n and vertical intercept $\log a$.



* We could also allow non-integer n; the term would then not strictly be polynomial, but we'd still say the function had "polynomial growth".

Case 2: Exponential → Linear

Suppose our original model was an exponential one:

$$y = ab^x$$

Then taking logs of both sides:

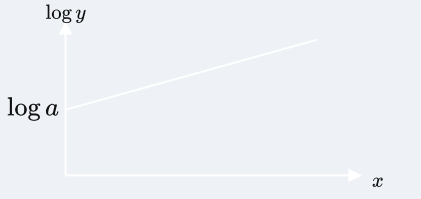
$\log y = \log ab^x$

$\log y = \log a + x \log b$

Again we can compare this against a straight line:

$$Y = mX + c$$

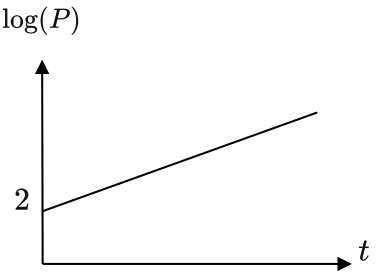
! If $y = ab^x$, then the graph of $\log y$ against x will be a straight line with gradient $\log b$ and vertical intercept $\log a$.



The key difference compared to Case 1 is that we're **only logging the y values** (e.g. number of transistors), not the x values (e.g. years elapsed). **Note that you do not need to memorise the contents of these boxes and we will work out from scratch each time...**

In summary, logging the y-axis **turns an exponential graph into a linear one**. Logging **both** the x and y-axis turns a polynomial graph into a linear one.

[Textbook] The graph represents the growth of a population of bacteria, P, over t hours. The graph has a gradient of 0.6 and meets the vertical axis at $(0, 2)$ as shown.



A scientist suggest that this growth can be modelled by the equation $P = ab^t$, where a and b are constants to be found.

a. Write down an equation for the line.

b. Using your answer to part (a) or otherwise, find the values of a and b, giving them to 3 sf where necessary.

c. Interpret the meaning of the constant a in this model.

a

b

c

[Textbook] The table below gives the rank (by size) and population of the UK's largest cities and districts (London is number 1 but has been excluded as an outlier).

City	B'ham	Leeds	Glasgow	Sheffield	Bradford
Rank, R	2	3	4	5	6
Population, P	1 000 000	730 000	620 000	530 000	480 000

The relationship between the rank and population can be modelled by the formula:
$P = aR^n$ where a and n are constants.

Textbook Error: They use $R = aP^n$ but then plot $\log P$ against $\log R$.

a) Draw a table giving values of $\log R$ and $\log P$ to 2dp.
b) Plot a graph of $\log R$ against $\log P$ using the values from your table and draw the line of best fit.
c) Use your graph to estimate the values of a and n to two significant figures.

a

b

c

A website wants to predict his number of Twitter followers P, t years from the start 2015. He predicts that his followers will increase exponentially according to the model $P = ab^t$, where a, b are constants that he wishes to find.

He records his followers at certain times. Here is the data:

Years t after 2015:	0.7	1.3	2.2
Followers P:	2353	3673	7162

a) Draw a table giving values of t and $\log P$ (to 3dp).
b) A line of best fit is drawn for the data in your new table, and it happens to go through the first data point above (where $t = 0.7$) and last (where $t = 2.2$). Determine the equation of this line of best fit. (The y-intercept is 3.147)
c) Hence, determine the values of a and b in the model.
d) Estimate how many followers the website will have at the start of 2020 (when $t = 5$).

Notes

1. We want to find the **parameters of a model**, e.g. $P = ab^t$ that **best fits the data** (in this case the parameters we want to find are a and b).
2. If the data had a linear trend, then this would be easy! We know from KS3 that we'd just plot the data, find the line of best fit, then use the gradient and y-intercept to work out the m and c in our linear model.
3. But the original data wasn't linear, and it would be harder to draw an 'exponential curve of best fit'.
4. We therefore log the model so that the plotted data then roughly forms a straight line, and then we can then draw a (straight) line of best fit.
5. The gradient and y-intercept of this line then allows us to estimate the parameters a and b in the original model that best fit the data.

The process of finding parameters in a model, that best fits the data, is known as **regression**.

a

t	
logP	

b

c

d

5 Kleiber's law is an empirical law in biology which connects the mass of an animal, m, to its resting metabolic rate, R. The law follows the form $R = am^b$, where a and b are constants.

The table below contains data on five animals.

Animal	Mouse	Guinea pig	Rabbit	Goat	Cow
Mass, m (kg)	0.030	0.408	4.19	34.6	650
Metabolic rate R (kcal per day)	4.2	32.3	195	760	7637

a Copy and complete this table giving values of $\log R$ and $\log m$ to 2 decimal places.

$\log m$	-1.52				
$\log R$	0.62	1.51	2.29	2.88	3.88

b Plot a graph of $\log R$ against $\log m$ using the values from your table and draw in a line of best fit.

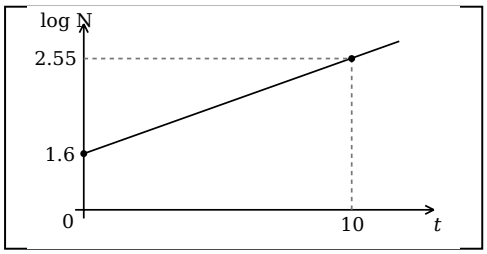
c Use your graph to estimate the values of a and b to two significant figures.

d Using your values of a and b, estimate the resting metabolic rate of a human male with a mass of 80 kg.

8 A scientist is modelling the number of people, N, who have fallen sick with a virus after t days.

From looking at this graph, the scientist suggests that the number of sick people can be modelled by the equation $N = ab^t$, where a and b are constants to be found.

The graph passes through the points $(0, 1.6)$ and $(10, 2.55)$.



a) Write down the equation of the line.
b) Using your answer to part **a** or otherwise, find the values of a and b, giving them to 2 significant figures.
c) Interpret the meaning of the constant a in this model.
d) Use your model to predict the number of sick people after 30 days.

Give one reason why this might be an overestimate.